White by Construction: Two Crypto Stat-Arbitrage Screens That Pass Their Own Placebos

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Abstract

We set out to build a profitable statistical-arbitrage strategy on hourly cryptocurrency pair spreads. We report mostly what does not work, and why that is the more useful result. Two screens that appear to find strong, tradable structure are statistical artifacts. A Kalman dynamic hedge ratio appears to produce stationary, mean-reverting spreads for nearly the entire liquid universe, but it runs a stationarity test on a filter's own innovations, which are white by construction; placebos that cannot be cointegrated pass the same screen at the same near-100% rate, while a clean test finds genuine out-of-sample cointegration in only a few percent of pairs. A rolling z-score appears to generate a large reversion edge, but it reverts mechanically even on a pure random walk. Both artifacts hold at hourly, four-hour, and daily frequencies. Once we benchmark against placebos, the structure that survives is either priced too fast to trade or too marginal to deploy: order-flow information in the Level-2 book decays within seconds, reversion speed is a selectable property of pairs (Spearman $\rho = 0.46$) but selectability is not profit, and the one market-neutral mean-reversion book that clears honest accounting posts a HAC- and venue-adjusted monthly Sharpe near 1.0, and only as a never-stop, multi-week hold; its one pre-registered arm gives a confidence interval that spans zero, so we read 1.0 as an exploratory best estimate, not a significant edge. We read the result as a market-efficiency statement about crypto pair spreads, and as a cautionary standard: a cointegration or z-score screen should not be trusted until it has been shown to fail on a matched placebo.

1 Introduction

We set out to build a profitable statistical-arbitrage strategy on hourly cryptocurrency pair spreads. We did not. The more useful finding is why: the two screens that made the strategy look best are statistical artifacts, and the structure that is genuinely present is either priced within seconds or too marginal to trade after costs and selection.

The project began in the pairs-trading tradition of Gatev, Goetzmann, and Rouwenhorst [10]: find two assets whose prices hold a stable relationship, trade the spread when it deviates, and collect the reversion. Cryptocurrency is a natural place to look, with a continuously traded market and a fast-changing set of listed coins. Our proposal was non-committal about profit from the start. It framed the goal as quantifying whether stable mean-reverting relationships exist in crypto, "as a way of gaining a deeper understanding of market efficiency and information flow." This paper answers that question, and the answer is mostly negative.

We make three contributions. First, two natural-looking screens manufacture the appearance of structure. A Kalman dynamic-hedge cointegration test reports out-of-sample cointegration in almost every liquid pair, but it tests a filter's own innovations, which are near-white by construction; placebos that cannot be cointegrated pass at the same rate the real pairs do, at hourly, four-hour, and daily frequencies. A rolling z-score reports strong mean reversion, yet it reverts mechanically even on a pure random walk. Each is exposed by a single matched placebo, and the placebo is the deliverable. Second, the structure that survives is not tradable: order flow in the Level-2 book carries real information with a seconds-scale half-life, and reversion speed is a selectable property of pairs that does not convert into profit. Third, one market-neutral mean-reversion strategy clears honest accounting at a monthly Sharpe near 1.0, but only without a stop and only as a patient multi-week hold, and a pre-registered held-out backtest of it inflates to a Sharpe near 5 on micro-cap moves no one could trade, so even the strategy that survives makes our point.

The methodological lesson is narrow and reusable: a cointegration or mean-reversion screen should not be believed until it has been shown to fail on a matched placebo. The backtest-overfitting literature [2, 12] checks the strategy that gets selected; we check the screen that builds the spread.

2 Related work

Our screens come from the pairs-trading and statistical-arbitrage literature. The distance and cointegration approaches of Gatev et al. [10] and Vidyamurthy [22], the residual mean-reversion framework of Avellaneda and Lee [1], and the taxonomy in Krauss [16] (with machine-learning variants in [17]) define what we test. Our dynamic hedge is the state-space pairs model of Elliott et al. [7], built on the cointegration of Engle and Granger [8]. The contribution here is not a new screen but a placebo that two standard ones fail.

For cryptocurrency the efficiency question is unsettled. Urquhart [21] finds early Bitcoin weak-form inefficient, while adaptive-markets evidence [20, 15, 19] finds efficiency time-varying and trending up. The closest prior on crypto pairs trading, Fil and Kristoufek [9], reports strategies that underperform benchmarks and are highly cost-sensitive. Our results agree and sharpen the mechanism: the apparent edge is either a screening artifact or priced within seconds.

The microstructure tests use standard tools: Kyle's λ [18], order-flow imbalance [4], VPIN [6], PIN [5], and information shares [13]. The placebo logic is the backtest-overfitting literature applied one level earlier. Bailey and coauthors show a handful of trials manufacture spurious in-sample performance [2, 3]; Harvey et al. [12] and Hou et al. [14] show most published factors fail multiple-testing correction; and Gelman and Loken [11] show data-contingent choices inflate error even without explicit fishing. We apply that skepticism to the screens that build the spread, not only to the strategy selected on it.

3 Data and market setting

The spread work uses hourly OHLCV for USDT spot pairs from the Binance.US API, stored locally, spanning 2021 through May 2026. The cross-exchange check pulls the same hours from Coinbase (USD-quoted) for the overlapping majors. The microstructure work uses event-level Level-2 data from Tardis: top-25 book snapshots and the trade tape for Binance.com (global) spot, the 50 most liquid USDT symbols, over 2024. We keep these separate on purpose, and note the venue gap: the spread strategy trades Binance.US, a much thinner venue, while the order-flow null is measured on the deeper global book. The spread and reversion results are Binance.US and Coinbase hourly klines; only the microstructure results use the Tardis book.

The investable universe is the 50 most liquid USDT symbols, $\binom{50}{2} = 1{,}225$ candidate pairs, constructed as of each analysis time. A symbol is eligible only if its history begins before the analysis window and reaches its start, a point-in-time rule added after an audit found that later listings could otherwise leak into the historical universe. The local store holds currently listed symbols, so delisted coins are mostly absent; we treat this as a survivorship limitation (Section 7), not a solved problem. Costs are charged on every position change at 15 bps per leg (10 bps fee, 5 bps slippage), about 30 bps round-trip for the two-leg spread, and stated before any performance number.

Three rules hold throughout. Inference uses Newey–West HAC standard errors, because overlapping targets and month-long holds make naive errors far too small; in one case a spurious five-minute reversal collapsed from $t = -22.8$ to $t = -1.4$ under HAC. Evaluation is strict walk-forward, six-month train and three-month test, with hedge ratios and pair selection fit on the train window only. And every positive claim is checked against a matched placebo, a random-walk, phase-randomized, block-shuffled, or random-pair null built to reproduce the claim if it is mechanical. Selection across many configurations is corrected with a deflated Sharpe ratio [2].

For the one trading result that survives (Section 6) we pre-registered a single configuration: the parameters and the held-out evaluation window were written to a file and committed before the run, then evaluated once and reported whatever the result was. The lock and its one pre-run correction are in the repository's commit history.

4 Two screening artifacts

Two screens drove our early results, and both are artifacts. The first, a Kalman dynamic hedge ratio, appeared to find out-of-sample cointegration in almost every liquid pair. The second, a rolling z-score, appeared to find strong mean reversion. We show that each passes a placebo it should fail, and that neither carries the information it seems to. This is the paper's main methodological contribution. The rolling z-score is a standard entry signal and the Kalman screen a natural way to test a drifting hedge, so both are easy to reach for and easy to be fooled by.

4.1 The Kalman dynamic-hedge cointegration screen

A static hedge ratio fits one β for the whole sample. A Kalman dynamic hedge lets β drift as a random walk and updates it each step from the data. We fit the process and observation noise by maximum likelihood on the training window, froze them, and rolled the filter forward on the test window. The test-period innovations, the filter's one-step prediction errors, became the spread, and we ran an augmented Dickey–Fuller (ADF) test on them. Under this screen 100% of the top-50 pairs rejected the unit root out of sample at $p < 0.05$, and 96.4% at $p < 0.001$. A static-OLS spread over the same window passed only about 3% of pairs. The contrast looked decisive.

It is decisive about the filter, not the pair. A Kalman filter with non-zero process noise adapts its hedge ratio to track whatever relationship is present. Its innovations are then driven toward whiteness, exactly white only for a correctly specified filter at steady state, and approximately so here on the frozen-parameter test window. An ADF test on a near-white series rejects the unit root almost always, whether or not the two assets are cointegrated. Testing the stationarity of a filter's own innovations is close to circular by construction, so the contribution is not that observation but the controls below, which show that even genuinely cointegrated pairs are indistinguishable from random walks under the screen. An earlier draft of this project defended the result on the grounds that the parameters were fit on training data only and never refit on test. That defense misses the mechanism. The whitening holds out of sample too, because the filter keeps adapting on the test window from its frozen dynamics.

A placebo settles it. We ran the identical screen on series that cannot be cointegrated: (a) independent simulated random walks, (b) each coin paired with a phase-randomized surrogate of another coin, and (c) real pairs with one leg block-shuffled. All three pass at the same near-100% rate as the real pairs (Table 1). A clean, non-circular test tells the opposite story. Engle–Granger [8] on the test window, and a static-OLS ADF, put genuine out-of-sample cointegration at 2–3% of pairs on the matched window, at or below the 5% null floor. The screen carries no cointegration information. We are not claiming that no pair ever co-moves, only that this screen cannot detect it.

Table 1: The Kalman screen passes placebos that cannot be cointegrated at the same rate as real pairs. Out-of-sample ADF on the Kalman innovations, top-50 USDT universe (250-pair subsample), $t_0 = 2024\text{-}01\text{-}01$, 90-day train / 30-day test.

Series tested	ADF $p < 0.05$	ADF $p < 0.001$
Real top-50 pairs	100.0%	96.4%
(a) Independent random walks	100.0%	98.3%
(b) Coin vs. phase-randomized surrogate	100.0%	95.0%
(c) Real pair, one leg block-shuffled	100.0%	95.0%
Clean tests (EG and static OLS, test window)	2–3%	—

We also ran a positive control: sixty constructed pairs that are genuinely cointegrated, $y_t = x_t + e_t$ with x_t a random walk and e_t a tight stationary AR(1). The Kalman screen passes them at 100%, the same rate it passes the random-walk negatives, so it does not separate true cointegration from none (Table 2). The mechanism is measurable: the test-window innovations are white, with a Ljung–Box test failing to reject whiteness for 87% of the cointegrated pairs and 95% of the random-walk pairs, while the static-OLS residual stays autocorrelated for both. The fitted hedge barely moves (median process variance $q_\beta \approx 2 \times 10^{-9}$), so the whitening is the filter fitting noise, not tracking a real relationship. The clean tests do separate the two, passing the positive control more often than the negative (17% versus 2% for Engle–Granger, 27% versus 8% for static-OLS), though genuine out-of-sample cointegration is hard to detect on a short window.

Table 2: Positive versus negative control. The Kalman screen passes both at the same rate and its innovations are white in both, so it carries no cointegration information; the clean tests separate them. 60 constructed pairs per group, 1500-bar train, 500-bar test.

	Positive (cointegrated)	Negative (random walks)
Kalman innovation ADF passes ($p < 0.05$)	100.0%	98.3%
Kalman innovations white (Ljung–Box $p > 0.05$)	86.7%	95.0%
Static-OLS residual white	0.0%	0.0%
Clean Engle–Granger passes	16.7%	1.7%
Clean static-OLS ADF passes	26.7%	8.3%

The artifact does not depend on the bar. We reran the screen at hourly, four-hour, and daily frequencies on a recent multi-year window (Table 3). Real pairs and random-walk placebos pass together at every cadence (Figure 1): 100% versus 100% at hourly and four-hour, and 70% versus 68% at daily ($p < 0.001$). The clean tests behave differently. Engle–Granger runs at 25% on this long hourly window and falls to the 5% null floor at daily; a static-OLS ADF runs higher still, from 42% at hourly to 8% at daily. Both are window-dependent. Over long hourly spans the crypto majors share a common trend, which a clean test reads as broad co-movement, and which the

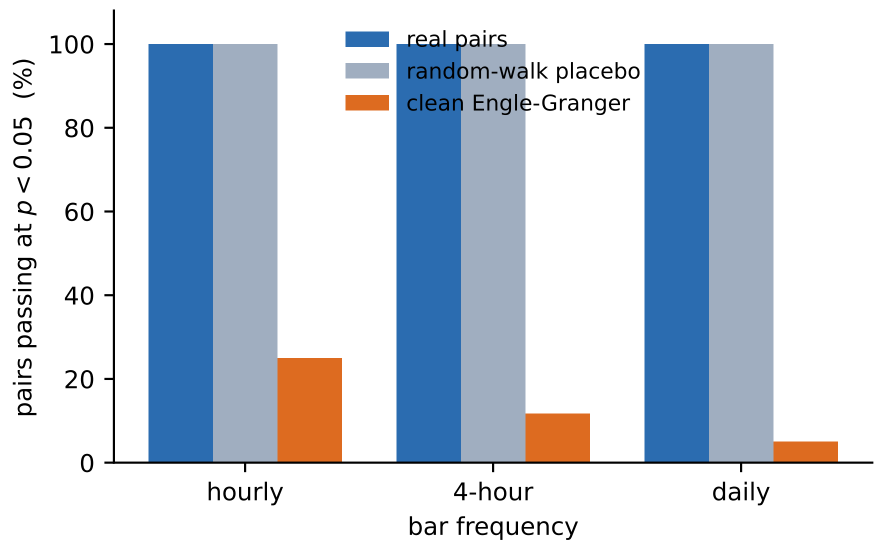

Figure 1: The Kalman screen carries no cointegration information at any frequency. Real pairs and random-walk placebos pass together at hourly, four-hour, and daily bars (near 100% except at daily, where both fall to about 70% under the stricter threshold), while a clean Engle–Granger test stays far lower. Exact values in Tables 1 and 3.

matched short window of Table 1 puts back at the 2–3% floor. Neither clean test approaches the screen's 100%, and the small difference between the two tables' real $p < 0.001$ rates (96% versus 100%) is window and sample size, not a contradiction. The Kalman screen carries no cointegration information at any frequency we tested.

Table 3: Frequency robustness of the Kalman artifact. Real pairs and random-walk placebos pass the screen together at every cadence; the clean Engle–Granger and static-OLS benchmarks do not. Top-50 universe, 60 pairs per cell, recent multi-year window.

	Real pairs		RW placebo		Clean tests $p < 0.05$	
Frequency	$p < 0.05$	$p < 0.001$	$p < 0.05$	$p < 0.001$	Engle–Granger	static OLS
Hourly	100.0%	100.0%	100.0%	100.0%	25.0%	41.7%
4-hour	100.0%	100.0%	100.0%	100.0%	11.7%	20.0%
Daily	100.0%	70.0%	100.0%	68.3%	5.0%	8.3%

4.2 The rolling z-score

The second screen is a rolling z-score, the standard entry signal for pair trades. For a spread s_t it subtracts a trailing mean and divides by a trailing standard deviation,

$$z_t = \frac{s_t - \bar{s}_{t-1}}{\sigma_{t-1}},$$

and it trades when $|z_t|$ is large, betting the spread returns to its mean. On our spreads it produced a large reversion edge.

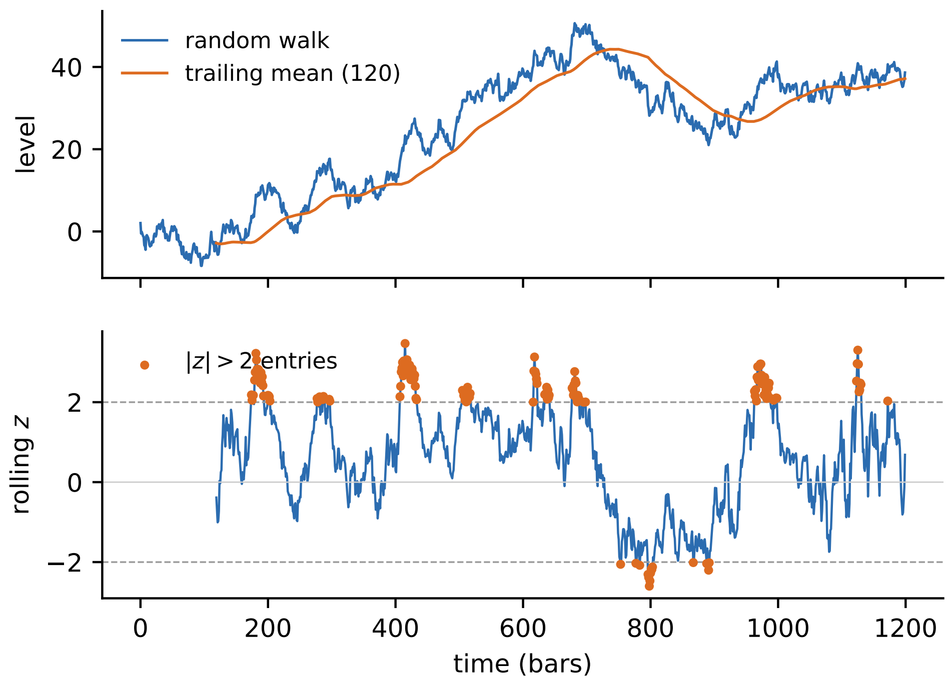

Figure 2: A rolling z-score manufactures mean reversion on a pure random walk. Top: one simulated random walk and its trailing mean. Bottom: the resulting rolling z, which crosses ± 2 and reverts toward zero, although the series has no mean to revert to.

The edge is mechanical. Subtracting a trailing mean from any series, including a pure random walk, manufactures mean reversion, because the trailing mean chases the level and the deviation from it cannot wander as freely as the raw series. Demeaning by a trailing window of length L induces negative autocorrelation of order $1/L$ in the result, a known bias, so the floor is calculable rather than fitted. A random walk has no mean to revert to, yet a rolling z-score on one still reverts.

We measured the mechanical floor two ways, and both are large. Per $|z| > 2$ event, the signed reversion on pure random walks is positive at every frequency we tested: $+0.98z$ at hourly, $+1.61z$ at four-hour, and $+1.51z$ at daily, each many standard deviations from zero (Table 4; Figure 2 shows the mechanism on one path). As a trading rule at hourly frequency, a rolling-z strategy that posts a Sharpe near 4.75 on our data still posts about 2.2 on random pairs and 2.4 on phase-randomized noise. Most of the apparent edge is the floor. For this reason every per-event reversion magnitude elsewhere in this paper is reported net of a per-pair, per-window random-walk floor.

Table 4: The rolling z-score manufactures mean reversion on pure random walks, at every frequency. Mean signed reversion per $|z| > 2$ event, 200 random-walk surrogates per cell.

Frequency	Mean event reversion (z)	s.d.
Hourly	+0.98	0.26
4-hour	+1.61	0.39
Daily	+1.51	0.45

Both screens share a failure mode. They manufacture the appearance of structure that a matched placebo reproduces. Both are robust to the analysis choices: the rolling-z floor stays positive ($+0.45$ to $+2.6z$) across lookback, horizon, and threshold, and the Kalman placebo passes at 85% or more across the ADF lag, so neither is an artifact of the parameters we picked. A cointegration screen built on a filter's own innovations must be checked against a white-noise placebo before it is believed, and a rolling-z reversion claim must be checked against a random-walk null. Neither check is standard practice, and without them both screens report a signal that is not there. This is the lesson the backtest-overfitting literature draws for strategy selection [2, 12], applied one level earlier, to the screens that build the spread.

5 What is real but not tradable

Strip the two artifacts and some genuine structure remains. None of it is tradable, for one of two reasons: it is priced faster than anyone could act, or it is a statistical property that does not convert into profit. We take the two clearest cases.

5.1 Reversion speed ranks pairs but does not pay

Mean reversion persists out of sample even though cointegration does not. Across nine disjoint walk-forward splits (six-month train, three-month test), a pair's in-sample OU reversion speed predicts its out-of-sample *excess* reversion, net of the rolling-z floor of Section 4.2: Spearman $\rho = 0.46$ (95% CI $[0.37, 0.54]$, $p < 0.001$), positive in every one of the nine splits. The top quintile shows about $0.69z$ more excess reversion than the bottom out of sample ($[0.58, 0.81]$, $p < 0.001$). The placebos are clean: a random quintile split gives $+0.0001z$, and shuffling the train-to-test link collapses ρ to -0.002. Reversion speed is a real, graded, persistent property that ranks pairs.

One caveat belongs on that coefficient. The $\rho = 0.46$ is the strongest of four in-sample reversion metrics we tried, chosen after the fact, and it is not corrected for that choice. The direction holds across all nine splits; the exact coefficient is a best case.

Selectability is not profit. Ranking pairs by reversion speed tells you where reversion lives. Whether it can be extracted after costs is a separate question, and the strategy built on the property (Section 6) is marginal at best.

5.2 Order flow is informative, but priced within seconds

We tested the order-flow ideas on a 2024 sample of Binance Level-2 book and trade data: aggregate book metrics on the top-50 USDT majors, and event-level tests on four majors (BTC, ETH, SOL, AVAX) across days drawn from all twelve months, from Tardis. The book carries real, contemporaneous information about price formation. It carries none at any horizon you could trade against, an efficiency null for both forecasting and execution.

Trade size does proxy information. Large trades carry about twice the one-second price impact of small ones, and the ranking replicates across all of 2024 on four coins (one-second impact for BTC of 0.33/0.23/0.13 bps for institutional, mid, and retail, with institutional 2.0 to 2.6× retail; the longer-horizon permanent estimate is noisier). Book-side order-flow imbalance explains within-second mid-price moves better than trade flow does (incremental R^2 of $+0.12$ to $+0.16$ for BTC and ETH), and 81 to 85% of the liquidity that leaves the best quote is cancellations, invisible to trade-only analysis.

All of it decays within seconds. The predictive content of order flow falls to near zero by 30 seconds, and at a 15-minute to one-hour horizon institutional flow predicts a mild reversal, not continuation, with an incremental R^2 near 0.03%. Hidden and iceberg liquidity is measurable (about 1 to 4% of at-best volume, with iceberg-episode rates higher) but carries no edge.

We then measured whether the signal helps execution, the usual fallback when a signal does not forecast returns. It does not (Table 5). Across 18,432 simulated parent orders over event-level book and tape, aggressive crossing is the cheapest method for these majors, a signal-timed passive rule does worse than a random one at the same posting rate, and the signal's correlation with the per-order advantage of posting over crossing is +0.06, noise. A perfect-foresight oracle would save 1.6 bps. The opportunity is real. No contemporaneous book feature forecasts it. The null holds across all twelve months of 2024, including the August 5 crash.

Table 5: Order-flow signals do not improve execution. Implementation shortfall versus arrival mid, pooled, 30-second horizon; lower is cheaper. 18,432 orders, BTC/ETH/SOL/AVAX.

Execution rule	Cost vs. arrival mid (bps)
Aggressive (always cross)	+1.34
Naive passive (always post)	+2.79
Signal-timed (post/cross on the L2 signal)	+1.98
Random at the same posting rate	+1.88
Oracle (perfect foresight)	−0.23

The one useful byproduct is a cost number. Walking the real book puts execution costs about 17 to 23% below the flat 5 bps slippage component the earlier backtests assumed. Cheaper costs do not rescue the strategy.

The efficiency reading is scoped to the liquid core. The Level-2 tests cover top-50 aggregate book metrics and four megacap symbols on the global venue; the thin Binance.US micro-caps where the marginal reversion edge of Section 6 lives are not in this sample. The two findings plausibly fit together, though across non-overlapping samples rather than on one venue: the majors we can measure are efficient, and the names with an apparent edge are too thin to trade. We did not measure the micro-caps' microstructure directly, so the join is an inference.

6 The marginal survivor

One strategy survives, but only barely, and its limits matter more than its headline. There is a real, market-neutral, diversified mean-reversion effect in crypto pair spreads. Its tradable magnitude is uncertain, it exists only without a stop, and a literal backtest of it inflates spectacularly. We lead with the bounds because they are the result.

6.1 The effect is real and market-neutral

Selecting pairs by reversion speed and holding the spread to convergence, with a static train-window hedge and z-score (not the rolling z of Section 4.2), produces a positive, market-neutral return. It is diversified: beta to an equal-weight crypto index is about −0.06, and the top ten principal-component portfolios explain only 1.7% of its return, so it is not a hidden factor bet. It replicates on an independent exchange (Coinbase, USD-quoted) and under four from-scratch reimplementations that share none of our pipeline, none of which found look-ahead, so the effect is not a coding artifact.

6.2 It exists only without a stop

The exit rule decides the sign of the P&L. With a conventional $|z| = 4$ stop the strategy loses (Sharpe -2.25), and the stop is the cause: it realizes losses on spreads that then revert. Remove the stop and hold to convergence and the strategy turns positive. The project set out to build a tightly risk-managed hourly strategy; this is the opposite, a patient never-stop book with a median hold near 35 days, in which 78% of positions do not converge within a quarter and about a third of the return is a generic survivor co-movement floor.

6.3 The magnitude is uncertain, and a literal backtest inflates it

The naive Binance Sharpe was first reported near 2.5, and four independent reimplementations put it at 2.0 to 2.2. Neither is the honest figure. Month-long holds make hourly returns heavily autocorrelated, so a Newey–West correction drops the monthly Binance number to about 1.4, and on the independent Coinbase venue it is about 0.85 to 0.90. The HAC- and venue-honest figure is near 1.0, and the edge concentrates in recent, high-volatility periods.

A pre-registered, held-out test makes the fragility concrete. We locked the configuration before running (Section 3 records the lock) and evaluated it once on held-out data, 2025-06 to 2026-05. The literal result was a monthly Sharpe near 5, *higher* than the in-sample figure, and it is not a tradable edge. It is broad-based inflation across high-volatility micro-cap alts: individual pair-windows post +300 to +520% in three months under a flat 15 bps cost that badly understates the true cost of trading those names, with unit-notional accounting that ignores capacity. A +517% three-month leg implies roughly a 150× gross return, not realizable at any size. Those names are barely traded: on Binance.US the held-out micro-caps (WAXP, ZIL, VTHO, ACH, EGLD) turn over a median of a few hundred to a few thousand dollars a day, roughly three to four thousand times less than the majors, so the deployable notional is negligible and a flat, size-independent cost is indefensible. The same machinery on a venue without those micro-caps returns the ∼1.0 figure. So the held-out test does not confirm an alpha; it shows the raw backtest is untrustworthy, the paper's thesis turned on our own surviving result.

The pre-registration had a second arm, which we report in full. A single selection over the whole pre-2025-06 history qualified only two pairs (long-horizon reversion in the tradeable band is rare), and they posted a monthly Sharpe of 1.60 with a HAC confidence interval of $[-0.21, 1.89]$ that spans zero. So the pre-registered test confirms the inflation and gives no significant evidence for a deployable edge. The ∼1.0 figure is itself exploratory, adjusted for HAC and venue after the search, and was never pre-registered.

6.4 What would have to be true

The effect is also selection-sensitive and survivorship-fragile. A deflated-Sharpe correction (Section 3) survives the no-stop family of configurations but fails the whole stop-versus-no-stop search, and the supports we might cite for it (the t-statistic, the $\rho = 0.46$ of Section 5, the market neutrality) share the same universe, selection, and windows, so they are correlated rather than independent confirmations. Its survivorship robustness rests on the selector avoiding coins that later delist, not on surviving them: a held position in a coin that collapses is a -100% leg. We tested this directly by forcing the book to hold through three real collapses (LUNA, UST, FTT, each paired with BTC, with the selector's avoidance overridden). Without a stop the loss runs -117% to -258% per pair, with intra-collapse drawdowns to $-1{,}300$%; the structural-break circuit breaker (halt after a single leg moves more than 50% against the position) caps the loss to about -65% to -91% and the worst drawdown to -117%. The breaker blunts the tail but does not remove it. At the book level, diversification absorbs individual breaks: a stress test that injects permanent breaks (one leg diverging up to 86% and never reverting) into a fraction of the 40 pairs each quarter erases the gross no-stop edge only near a 20%-per-quarter ($\sim$80%/yr) attrition rate, far above realized delisting frequencies. That bounds the mechanism rather than the HAC-honest Sharpe, but it shows the per-pair tail does not by itself drive the book. A fully point-in-time universe of every delisted coin is still beyond our data.

The defensible claim is therefore the sensitivity itself. There is a real, market-neutral reversion signal in crypto pair spreads that can be selected by reversion speed. Whether it is worth anything after realistic costs, capacity, and selection is unresolved, and the most honest single number, a monthly Sharpe near 1.0, is an estimate that a literal backtest does not cap from above.

7 Limitations

Five limitations bound these results, and the survivorship gap is the one most likely to change the conclusion.

- **Survivorship.** The local store holds currently listed symbols, so most delisted coins are absent. The surviving strategy's robustness rests on the selector avoiding coins that later delist; forced to hold through three real collapses (Section 6) the no-stop loss runs -117% to -258% per pair and the circuit breaker caps it near -65% to -91%. A complete point-in-time universe of every delisted coin is still beyond our data.
- **Selection.** No single configuration was fixed before the search and evaluated once across the whole project. The pre-registered test of Section 6 covers one configuration but cannot undo the selection that produced it, so the deflated-Sharpe caution stands.
- **Microstructure coverage.** The event-level Level-2 tests use four symbols (BTC, ETH, SOL, AVAX) over days sampled across 2024, not a continuous year on the full universe. The aggregate book metrics use the top-50; the execution and impact results are four-symbol.
- **Single venue and quote.** The spread results are Binance.US, USDT-quoted, with a Coinbase cross-check; the microstructure is Binance spot. We have not tested other quote currencies or a broader set of venues.
- **Costs and capacity.** Costs are a flat per-leg charge. We do not model market impact or capacity, which is exactly what inflates the held-out micro-cap backtest of Section 6.

8 Conclusion

Our negative results come from the screens we used, not from the market. The Kalman cointegration test and the rolling z-score both pass placebos they should fail, at every frequency we tested. Order flow forecasts only the next few seconds, a horizon no one can trade. The one market-neutral reversion effect that survives is real but marginal, stop-sensitive, and selection-sensitive, and a literal backtest of it inflates on costs it does not model.

The practical lesson is a standard, not a strategy. A cointegration or mean-reversion screen should be trusted only after it has been shown to fail on a matched placebo built to reproduce the claim if the claim is mechanical. For crypto the evidence points to microstructure efficiency in the liquid majors at tradeable horizons: the information is real, and the market has already priced it.

What would move the result is a genuinely point-in-time universe with delisted coins, a capacity- and impact-aware cost model, and a single pre-registered configuration evaluated once on data the search never saw. Until then, the efficiency reading holds for the liquid core we can measure; for the thin tail where delisting risk lives, we say undetermined, not efficient.

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